

MT CARMEL, IL, USA

WMO: 720330

Lat: 38.607N

Lon: 87.727W

Elev: 161

StdP: 99.40

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63853

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-11.2	-19.2	0.7	-12.4	-16.4	0.9	-10.3	12.6	4.8	11.5	3.2	2.4	320	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.8	23.9	32.6	23.5	31.8	23.2	26.4	31.0	25.6	30.2	24.9	29.4	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.7	29.2	24.1	19.4	28.3	23.3	18.4	27.4	83.1	30.6	79.8	30.1	76.1	29.6	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.1	8.1	DB	-18.7	35.8	2.6	1.9	-20.5	37.1	-22.1	38.2	-23.5	39.3	-25.4	40.6	(4)
(5)			WB	-19.1	27.6	2.5	0.9	-20.9	28.2	-22.4	28.7	-23.9	29.2	-25.7	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	0.0	1.7	7.6	13.7	19.5	24.1	24.6	23.8	20.7	14.1	7.3	2.4
	DBStd	10.14	6.35	6.14	6.05	5.18	4.73	2.89	2.84	2.88	3.77	5.16	5.18	5.67
	HDD10.0	1071	316	239	121	23	1	0	0	0	0	17	112	243
	HDD18.3	2574	570	465	337	153	46	1	0	2	18	154	333	495
	CDD10.0	2292	4	8	46	135	295	422	454	427	321	143	30	7
	CDD18.3	751	0	0	3	14	82	173	196	170	89	22	1	0
	CDH23.3	7317	0	0	26	148	752	1679	1950	1592	936	229	4	0
	CDH26.7	2784	0	0	3	21	255	675	787	613	363	66	1	0
Wind	WSAvg	3.2	4.1	3.9	4.1	4.0	3.3	2.8	2.0	1.8	2.2	3.1	3.4	3.7
Precipitation	PrecAvg	1119	87	65	90	109	133	118	92	82	94	79	90	86
	PrecMax	1481	178	130	266	289	288	234	174	211	204	234	185	189
	PrecMin	758	33	15	21	29	39	5	24	22	7	0	0	34
	PrecStd	192	39	30	55	60	67	51	43	47	52	59	48	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.5	20.5	27.1	28.6	32.8	35.2	37.0	35.1	34.8	31.4	23.2	18.2
		MCWB	14.9	15.5	17.8	18.0	21.2	23.0	25.1	24.0	22.1	20.7	16.3	15.0
	2%	DB	15.0	17.4	22.7	27.2	31.2	33.0	33.2	32.9	32.5	28.3	21.1	16.1
		MCWB	12.9	13.4	16.1	17.4	20.4	23.4	25.2	23.9	22.1	19.7	15.4	13.9
	5%	DB	12.4	14.4	20.2	25.1	29.8	32.2	32.3	32.0	31.0	26.1	18.0	12.9
		MCWB	10.5	11.8	14.5	16.6	20.3	22.9	25.0	24.0	22.1	18.0	13.8	10.8
	10%	DB	9.1	11.3	17.5	22.7	27.7	30.9	31.0	30.2	28.6	23.0	16.1	11.0
		MCWB	7.2	8.4	13.3	15.6	19.6	22.3	24.2	23.4	21.0	17.0	11.8	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	16.7	18.5	20.8	23.9	26.3	27.8	27.5	25.1	22.9	18.9	16.5
		MCDB	16.8	19.5	23.4	26.5	28.8	31.3	32.5	31.8	30.7	28.3	21.7	18.2
	2%	WB	13.3	14.5	17.2	19.3	22.9	25.4	26.8	26.1	24.0	21.5	16.8	14.2
		MCDB	15.0	16.9	22.2	24.2	28.1	30.5	31.3	30.5	29.4	26.2	19.5	15.9
	5%	WB	10.8	12.0	15.6	18.0	21.9	24.5	25.9	25.3	23.1	19.9	14.6	11.5
		MCDB	11.9	14.5	19.3	22.2	27.0	29.5	30.4	29.9	28.5	24.1	17.8	12.8
	10%	WB	7.0	8.3	13.7	16.9	21.0	23.7	25.2	24.4	22.2	18.1	12.2	8.8
		MCDB	8.6	10.7	17.0	21.1	25.5	28.5	29.5	28.5	26.7	22.0	15.2	10.4
Mean Daily Temperature Range	5% DB	MDBR	8.3	8.9	10.0	11.6	10.9	11.0	10.8	11.5	13.1	12.2	10.2	7.8
		MCDBR	11.2	13.1	13.5	14.2	12.9	12.4	11.8	13.1	15.1	14.8	13.6	10.5
		MCWBR	9.6	10.3	8.0	7.0	4.7	4.8	5.0	5.7	5.9	7.1	9.0	9.1
	5% WB	MCDBR	10.6	12.2	12.1	11.0	10.9	10.5	10.1	10.9	12.3	12.5	11.6	9.9
		MCWBR	10.0	10.5	8.0	6.8	4.9	4.7	5.1	5.6	5.7	6.8	9.2	9.4
Clear-Sky Solar Irradiance	taub	0.311	0.323	0.359	0.405	0.442	0.446	0.466	0.472	0.418	0.360	0.322	0.308	
	taud	2.452	2.399	2.366	2.240	2.189	2.205	2.155	2.123	2.266	2.427	2.480	2.466	
	Ebn at Noon	866	902	899	872	844	838	817	801	826	848	846	843	
	Edh at Noon	82	98	112	136	145	143	149	151	123	95	79	75	
All-Sky Solar Radiation	RadAvg	1.92	2.68	3.72	4.90	5.51	6.36	6.15	5.62	4.66	3.37	2.27	1.64	
	RadStd	0.21	0.29	0.34	0.28	0.48	0.41	0.39	0.31	0.33	0.36	0.25	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	+75

Nomenclature: See separate page